GABRIEL CORTEZ

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Extended version of the portfolio · gabrielcortezspr

$ SUMMARY

Software Engineer focused on automation, data and AI at Samsung Electronics (Visual Display Division), building internal software and automation for TV production engineering, from the initial idea and requirements to deployment and adoption. I co-founded a conversational-AI startup with agents in production, build an open-source AI agent orchestrator backed by my own comparative research, and have worked in embedded-software QA, FPGA research and test automation. Computer Engineering student and medalist in international and national science olympiads.

$ SKILLS

Languages: Python, SQL, TypeScript, JavaScript, Java, C/C++, Rust (project), Verilog

Data: pandas, NumPy, Matplotlib, ETL, real-time pipelines, data quality, dashboards

ML & vision: scikit-learn, model training, validation and retraining (data drift), YOLO, OpenCV

Evaluation & QA: benchmarks with hidden tests, regression, adversarial testing, hallucination detection, JUnit, Postman, BDD/Cucumber

Tools & methods: Git, GitHub, Claude Code, Codex, Tauri · Scrum, Kanban

Back-end & web: FastAPI, Flask, REST APIs, SQLAlchemy, Supabase, React, Next.js

AI / LLM: LLMs, RAG, embeddings, NLP, agents and multi-agent orchestration, prompt engineering

Automation: RPA, Selenium, PyAutoGUI, BeautifulSoup, n8n

Embedded: FPGA (Cyclone IV), ESP32, MQTT, Arduino, Raspberry Pi, Android (testing and logs)

Languages spoken: Portuguese (native), English (advanced, daily use with a Korean team)

$ EXPERIENCE

Software Engineer — Feb 2026 – Present
Samsung Electronics, Visual Display Division – Manaus, Brazil

- Internal software and automation for TV production engineering, from idea and requirements to deployment and adoption.
- Built real-time data pipelines processing high-volume production line telemetry, replacing manual collection with automated ingestion and transformation.
- Shipped dashboards used by leadership in Brazil and Korea to monitor process KPIs.
- Developed internal Python tooling adopted by every engineer on the team, standardizing previously improvised workflows.
- Designed and deployed RPA automations for legacy systems with no API (Selenium, PyAutoGUI, BeautifulSoup, OpenCV).
- Trained and deployed machine learning models (scikit-learn, pandas, NumPy) and applied AI-driven analysis to factory data.
- Drove technical direction independently across architecture, requirements and deployment on a team with no dedicated software engineering function.

Co-founder & AI Developer — Oct 2025 – Feb 2026
Kacto – Manaus, Brazil

- GPT-4o conversational agent for automated customer service at clinics, in production at 3 clinics. Startup from the Ocean Launch program.
- RAG pipeline: knowledge-base chunking, embeddings and vector indexing on Supabase.
- Prompt engineering and evaluation: regression tests, adversarial testing, hallucination detection and guardrails; workflow automation with n8n and Python.

QA Intern — Feb 2025 – Feb 2026
Sidia Institute of Science and Technology – Manaus, Brazil

- Embedded software QA on Samsung Galaxy devices in a global program: functional, integration and regression testing, manual and automated.
- Validated firmware, OS and app integration and regulatory behaviors; investigated failures through device logs with the engineering team in Korea.

Embedded Systems Research Fellow (FAPEAM) — Aug 2024 – Jul 2025
SENAI – Manaus, Brazil

- Automated greenhouse on an FPGA (Cyclone IV) in Verilog: sensors, automation, environmental monitoring and hardware-software communication. Presented at the international scientific event Chip in the Jungle.

Test Developer (Fellow) Sep 2024 – Dec 2024
Fundação Muraki – Manaus, Brazil

- Automated tests in Java with Selenium and JUnit, API testing in Postman and BDD scenarios with Cucumber in an agile team.

Freelance Web Development & Design Aug 2026 – Present
Manaus, Brazil

- Websites for companies and professionals such as Grupo Shamar (security services) and Horizont Rec (video production); graphics for event companies.

$ PROJECTS € RESEARCH

NarraTer · open-source AI agent orchestrator 2026 – Present
Tauri 2, Rust, React, TypeScript

- Linux app where terminals running agents from different vendors (Claude Code, Codex, Gemini CLI) become nodes on an infinite canvas; edges define who can talk to whom.
- Agent-to-agent protocol that needs no cooperation from the agent: delivery through the terminal, end of turn read from the screen and transcript.
- Own benchmark comparing 7 protocol versions: premature answers down to 0 of 40 and cost per task from US$ 1.02 to US$ 0.65.
- "Team or solo agent" study: on a sequential task, a single agent was 2.5× faster and used 7× fewer tokens with the same score.

Pro-Link Challenge (CREA-AM, II CENATEC) · winning team lead Sep 2026
PHP, Docker, MySQL

- Marketplace connecting registered engineers to companies through an evidence engine over technical records (ART/CAT), auditable random tie-breaking and LGPD built into the product.

BusFlow · IoT passenger counting 2025
ESP32, MQTT, Node.js

- ESP32 firmware with proximity sensors publishing data over MQTT. State finalist at Liga Jovem 2025.

Tabletop · chess and Go in pixel art 2026
Python, OpenGL, GLSL

- Chess and Go table with Balatro-style shaders, full official rules and Chinese/Japanese scoring for Go.

Exploratory data analysis (UEA) 2026
Python, pandas, Matplotlib

- Two Kaggle datasets (Netflix, 8,807 titles; 1,000 students) with original questions crossing variables.

$ AWARDS € OLYMPIADS

International

- Bronze · American Mathematics Olympiad (AMO), 2021
- Bronze · Singapore Math Olympiad, 2021
- Bronze · Mathematics Without Borders, 2021

National & state

- Gold · Brazilian Astronomy Olympiad (OBA), 2020
- Silver · National Science Olympiad (ONC), 2019
- Honorable Mention · ONC, 2021
- Silver · Brazilian Financial Education Olympiad, 2020
- Silver (2020) and Bronze (2021) · Brazilian Rocket Show (MOBFOG)
- Bronze · Amazonas Chemistry Olympiad, 2020

Competitions & events

- Winner of the Pro-Link Challenge (CREA-AM), 2026
- State finalist · Liga Jovem (BusFlow), 2025
- Presenter at Chip in the Jungle, 2025

$ EDUCATION

Amazonas State University (UEA) · B.Sc. in Computer Engineering Jun 2022 – Dec 2027
Additional: Ocean Launch (startup pre-acceleration), CI Amazônia (microelectronics), embedded systems at FPF Tech, STEM Academy.